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OSSP:

Commands:

Uname : Displays system information.

Lscpu : Shows CPU details.

```
Last login: Fri Aug 7 15:00:56 on ttys000
sreenikhilreddy@Srees-MacBook-Pro: ~ % ls
1.c.save      Documents
12            Downloads
123.c.save    Downloads
2            Library
2.c           Music
3            myFolder
3.c           NewFolderName
4            Parallels
4.c           Pictures
4.c.save     PROJECT DSA 2
a.out        projects
Applications  Public
Applications (Parallels)  python
Desktop      task
dir1         task.c
dir2         terminal
dir3
```

ps : Displays the currently running processes.

```
Last login: Fri Aug 7 15:00:56 on ttys000
sreenikhilreddy@Srees-MacBook-Pro: ~ % ls
1.c.save      Documents
12            Downloads
123.c.save    Downloads
2            Library
2.c           Music
3            myFolder
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4.c.save     PROJECT DSA 2
a.out        projects
Applications  Public
Applications (Parallels)  python
Desktop      task
dir1         task.c
dir2         terminal
dir3

sreenikhilreddy@Srees-MacBook-Pro: ~ % pw
zsh: command not found: pw

sreenikhilreddy@Srees-MacBook-Pro: ~ % uname
Darwin

sreenikhilreddy@Srees-MacBook-Pro: ~ % ps
  PID TTY          TIME CMD
 1311 ttys000    0:00.04 -zsh
sreenikhilreddy@Srees-MacBook-Pro: ~ %
```

Relation Between Hardware Resources and Operating System Services:

1. CPU

The operating system shares the CPU among different programs so they can run together.

Example: Chrome and a music player both receive CPU time.

2. Memory (RAM) → Memory Management

The operating system provides separate memory space for each running program.

Example: Chrome and VS Code use their own memory.

3. Storage (SSD) → File System Management

The operating system manages and organizes files on the storage device.

Example: Saving a file in the **Documents** folder.

4. I/O Devices → Device Management

The operating system controls input and output devices using device drivers.

Example: Using a printer or keyboard through the operating system.